

Anomalous Magnetic Moments in FFGFT Theory

Geometric Derivation from Time-Mass Duality

Purely Geometric Formulas and Precise Ratio Predictions

Revision 11

Johann Pascher

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Correction notes

Corrections from the calculation review (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note there with the previous value:

- $a_\tau = 1.277 \times 10^{-3}$: note (rounded measured values; exactly 1.282×10^{-3} , Rev. 12, Doc. 158) – a difference in precision, not a calculation error.
- Literature examples for the exponents: $p_\mu = 5/3$ now the Flory value of the self-avoiding walk in 3D (previously 2D Brownian motion, random walk, Koch curve); $p_\tau = 4/3$ now the frontier of planar Brownian motion and the planar self-avoiding walk (previously Koch curve, Mandelbrot set). The exponents themselves are settings.
- Effective projection factor: $k_{\text{eff}} = 2.2692 \rightarrow 2.2688$ (previously divided with the rounded $k_{\text{geom}} = 2.224$ instead of 2.22357); distance to k_{rec} : $0.014 \rightarrow 0.033\%$. Consequential values: $a_e^{(\text{corr})} = 1.1600$, $a_\mu^{(\text{corr})} = 1.1644$ (0.13 %), $a_\tau^{(\text{corr})} = 1.2456 \times 10^{-3}$ (rounded 1.246); ideal deviation $2.03 \rightarrow 2.07\%$; improvement factors $145 \rightarrow 62$ and $13 \rightarrow 15$ – at all places.
- k_{rec} : denominator $1.160 \rightarrow 1.15965 \times 10^{-3}$ (only then does 2.2696 follow).

Abstract

In the present work, the fundamental architecture of spacetime is reinterpreted within the framework of **Fundamental Fractal Geometric Field Theory (FFGFT)** – internally referred to as the T0 model (B18). The central idea is the transition from a point-like to a purely geometric description of the vacuum as a four-dimensional **Gyral Torus**.

Note on versions. The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

Geometric Structure: The theory is based on the fractal-geometric foundation with the parameter $\xi \approx (4/3) \times 10^{-4}$ and the densest local sphere packing by regular **Tetrahedra**. This tetrahedral basis forms the stable foundation for the low generations (electron, muon, proton/neutron) as well as the local 3D crystal structure of the torus. Building upon this, the ideal sub-Planck factor

$$f = 7500,$$

emerges through fractal branching and pentagonal symmetry breaking, representing an exactly 7500-fold reduction compared to the conventional Planck scale (t_0) and following directly from the geometric winding density $30000/4$.

g-2 Anomaly: A core element of the work is the transparent geometric derivation of the anomalous magnetic moments of leptons. While the Standard Model relies on numerous perturbative terms, in FFGFT the electron anomaly follows directly from the base winding (tetrahedral projection). The muon and tau anomalies arise from fractal branchings with Hausdorff dimensions $p \approx 5/3$ and $4/3$, respectively. With the ideal value $f = 7500$ and the geometric projection factor $k_{\text{geom}} = (2/\sqrt{\varphi})\sqrt{2} = 2.224$, the purely geometric predictions achieve an accuracy of about 2 %. Through the **fractal correction** $K_{\text{frak}}^{3/2}$, this systematic deviation is captured geometrically (the exponent $3/2$ is a motivated choice): the effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2688$ agrees with the experimentally reconstructed value $k_{\text{rec}} = 2.2696$ to 0.03 %. The corrected absolute values achieve 0.03% (electron) and 0.13% (muon) accuracy. The most precise, k_{geom} -independent prediction for the tau anomaly is

$$a_\tau \approx 1.277 \times 10^{-3},$$

which follows exclusively from the exact ratio $f^{1/3} - 1$.

Geometric Proportionality: All physical base quantities (constants, masses, couplings) stand in fixed geometric ratios, considerably reducing the number of free parameters compared to the Standard Model. The T0 theory thus offers a geometric description and provides concrete, experimentally testable predictions – particularly for the tau anomaly as an important test at Belle II.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

Note on Older Documents

Previous versions of the g-2 analysis (018_T0_Anomale-g2-9/10) used semi-empirical factors or the ideal k_{geom} without fractal correction. The present Rev. 11 shows that the 2 % deviation of absolute values **can be traced back to the fractal correction** $K_{\text{frak}}^{3/2}$ – the effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$ agrees with the experimentally reconstructed value to 0.03 %.

Keywords: Anomalous magnetic moment, g-2, T0 theory, Time-Mass Duality, Torsion lattice, Ratio predictions, Koide formula

.1 Introduction: Geometric vs. Semi-Empirical Approaches

The Philosophy of T0 Theory

The T0 theory is based on the principle that **all** physical constants should follow from the geometric structure of a 4-dimensional torsion lattice. For anomalous magnetic moments this means:

- **No** hidden fit parameters
- **Only** geometric factors: φ, ξ, f
- Open statement of precision limits
- Consistency with other predictions

Consistency with Mass Predictions

The T0 theory predicts lepton masses with 0.4–1.2% deviation:

Lepton	T0 [MeV]	Exp [MeV]	Deviation
Electron	0.505	0.511	1.18%
Muon	105.0	105.7	0.66%
Tau	1783	1777	0.37%

Table 1: Lepton masses in T0

Expectation: g-2 should have similar precision (1%).

With the fractal correction $K_{\text{frak}}^{3/2}$, the g-2 predictions achieve 0.03–0.13% accuracy, comparable to the mass precision.

.2 Physical Fundamentals

What is the Anomalous Magnetic Moment?

The magnetic moment of a charged spin-1/2 particle is:

$$\mu = g \cdot \frac{e}{2m} \cdot \frac{\hbar}{2} \quad (1)$$

where g is the gyromagnetic factor (g-factor).

Dirac Prediction: For a point-like particle: $g = 2$

Quantum Effects: Vacuum polarization, vertex corrections $\Rightarrow g \neq 2$

Anomaly: $a = (g - 2)/2$

QED Expectation: $a \approx \alpha/(2\pi) + \mathcal{O}(\alpha^2) \approx 0.00116$

T0 Interpretation: Windings in the Torsion Lattice

In T0 theory, leptons are **winding structures** in the 4D torsion lattice:

- **Electron:** Simple winding (1st generation)
- **Muon:** Winding with fractal branching (2nd generation)
- **Tau:** More complex fractal structure (3rd generation)

The anomalous moment arises from:

1. The **rotation** of the winding (spin)
2. The **charge distribution** on the winding
3. The **projection** 4D $\rightarrow$ 3D
 $\Rightarrow$ **No** point-like charge $\Rightarrow a \neq 0$

.3 Geometric Formulas

Fundamental Parameters

The T0 theory uses exclusively three geometric fundamental constants:

$$\varphi = \frac{1 + \sqrt{5}}{2} = 1.618 \dots \quad (\text{Golden Ratio}) \quad (2)$$

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \times 10^{-4} \quad (\text{Torsion constant}) \quad (3)$$

$$f = 7500 \quad (\text{Sub-Planck factor}) \quad (4)$$

The Real Sub-Planck Factor: $f = 7500$

Now we put everything together: The ideal crystal remains intact, the symmetry breaking only affects the projection factors:

$$\boxed{f = 7500} \quad (5)$$

This is a **central number of the T0 theory**. It appears in almost all formulas and describes:

- The number of Sub-Planck cells per Planck length
- The density of the torsion lattice
- The fundamental frequency of all geometric resonances

The Symmetry Breaking: The Role of the Golden Ratio

A perfect, ideal crystal would be completely symmetric. Yet our world shows symmetry breaking on all levels:

- Matter dominates over antimatter
- The weak interaction violates parity symmetry
- The neutron is heavier than the proton
- The three lepton generations have different masses

In the T0 theory, all these symmetry breakings have a single, geometric origin: the pentagonal symmetry of the crystal, embodied by the **golden ratio** φ . The golden ratio $\varphi = (1 + \sqrt{5})/2 = 1.618033989 \dots$ is the irrational number describing pentagonal symmetry. In a perfect pentagon, φ appears everywhere: The ratio of diagonal to side is exactly φ . Why pentagonal symmetry specifically? For deep mathematical reasons, pentagonal symmetry is the first one that **cannot tile the plane periodically**. This leads to *quasicrystals* – structures that are ordered but not periodic. Exactly such a quasicrystalline structure is postulated by T0 theory for the Sub-Planck scale. The symmetry breaking is quantified in the theory not by a direct subtraction of 5φ from the ideal anchor number 7500. Instead, it manifests in the **fractal correction** $K_{\text{frak}}^{3/2}$, which corrects the ideal geometric projection factor k_{geom} to the effective value k_{eff} :

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} \approx 2.22357, \quad (6)$$

which projects the 4D torsion onto the 3D world. The **fractal correction** yields the effective projection factor:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.2236}{0.98007} = 2.2688 \quad (7)$$

The exponent $3/2$ corresponds to the half spatial dimension ($D/2 = 3/2$) and describes how the fractal structure $K_{\text{frak}} = 1 - 100\xi = 0.98667$ acts across three spatial dimensions on the projection.

Essential: The experimentally reconstructed value $k_{\text{geom}}^{\text{rec}} = 2.26955$ agrees with $k_{\text{eff}} = 2.2688$ to **0.033%** – the 2% deviation of the ideal formula can thus be traced back to the fractal correction. *(Corrected on 30 Sept. 2026: previously $2.224/0.98007 = 2.2692$ and 0.014% – the value $k_{\text{geom}} = 2.224$ rounded to four digits had been inserted; with $k_{\text{geom}} = 2.22357$, $k_{\text{eff}} = 2.2688$ follows and a distance of 0.033% from $k_{\text{rec}} = 2.2696$. Likewise at all subsequent places.)*

From the ideal 7500 remained the ideal 7500. In the T0 model this number becomes the fundamental constant. It determined how densely the lattice was packed, how quickly torsion could propagate, which resonances were possible. In the T0 picture the observed particle masses and force strengths are meant to go back to this single geometric story, and likewise – **[S]**, the cosmic sector is deliberately set aside in the current corpus (P39) – the cosmological constants: From perfect crystal to pentagonally broken reality, with the breaking quantitatively captured by $K_{\text{frak}}^{3/2}$.

Electron: Base Winding

Formula:

$$a_e = \frac{S_3/f}{k_{\text{geom}}} \quad (8)$$

where:

- $S_3 = 2\pi^2 = 19.739$: 3D surface of the 4D winding
- $f = 7500$: Sub-Planck scaling
- k_{geom} : Geometric projection factor

Geometric Projection Factor:

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} \quad (9)$$

Explanation of Factors:

- $2/\sqrt{\varphi} = 1.572$: Pentagonal projection (from ξ -structure)
- $\sqrt{2} = 1.414$: Diagonal projection 4D $\rightarrow$ 3D
- $k_{\text{geom}} = 2.224$: determined purely geometrically

Numerical Calculation:

$$k_{\text{geom}} = \frac{2}{\sqrt{1.618}} \times \sqrt{2} = 2.224 \quad (10)$$

$$a_e = \frac{19.739/7500}{2.224} \quad (11)$$

$$a_e = 1.184 \times 10^{-3} \quad (12)$$

Comparison (ideal zeroth order):

- T0 (ideal): $a_e = 1.184 \times 10^{-3}$
- Experiment: $a_e = 1.160 \times 10^{-3}$
- Deviation: **2.07%**

With fractal correction $K_{\text{frak}}^{3/2}$:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.2236}{0.98007} = 2.2688 \quad (13)$$

$$a_e^{(\text{corr})} = \frac{S_3/f}{k_{\text{eff}}} = \frac{19.739/7500}{2.2688} = 1.1600 \times 10^{-3} \quad (14)$$

- T0 (corrected): $a_e = 1.1600 \times 10^{-3}$
- Experiment: $a_e = 1.1597 \times 10^{-3}$
- Deviation: **0.033%**

(Corrected on 30 Sept. 2026: previously $a_e^{(\text{corr})} = 1.1598$, $a_\mu^{(\text{corr})} = 1.1642$, $a_\tau^{(\text{corr})} = 1.2454 \times 10^{-3}$ with 0.014 % and 0.15 % deviation – consequence of $k_{\text{eff}} = 2.2688$: $a_e = 1.1600$ (+0.033 %), $a_\mu = 1.1644$ (−0.13 %), $a_\tau = 1.2456 \times 10^{-3}$. The ideal deviation is 2.07 % (previously 2.03 %). Likewise in all tables and summaries.)

Muon: Fractal Additional Winding

Formula:

$$a_\mu = a_e + \Delta a_{\text{fractal}} \quad (15)$$

with

$$\Delta a_{\text{fractal}} = \frac{4\pi}{f^{p_\mu}} \quad (16)$$

where:

- $p_\mu = 5/3$: Fractal Hausdorff dimension

- 4π : Complete torsion revolution

Meaning of $p_\mu = 5/3$:

This is the well-known Hausdorff dimension of:

- the self-avoiding random walk in 3D (Flory value $1/\nu = 5/3$) (*Corrected on 30 Sept. 2026: previously "Brownian motion in 2D, self-avoiding random walk, Koch curve" – misattributed: planar Brownian motion has dimension 2 (its frontier $4/3$), the planar self-avoiding walk $4/3$, the Koch curve $\log 4 / \log 3 = 1.262$; $5/3$ is the Flory value of the self-avoiding walk in 3D. The exponent itself is a setting and is not affected.*)
 $\Rightarrow$ Physically plausible for a "partially branched winding".

Numerical Calculation:

$$\Delta a_{\text{fractal}} = \frac{4\pi}{7500^{5/3}} = 4.373 \times 10^{-6} \quad (17)$$

$$a_\mu = 1.184 \times 10^{-3} + 4.373 \times 10^{-6} \quad (18)$$

$$a_\mu = 1.188 \times 10^{-3} \quad (19)$$

Comparison (ideal zeroth order):

- T0 (ideal): $a_\mu = 1.188 \times 10^{-3}$
- Experiment: $a_\mu = 1.166 \times 10^{-3}$
- Deviation: **1.89%**

With fractal correction:

$$a_\mu^{(\text{corr})} = a_e^{(\text{corr})} + \Delta a_{\text{fractal}} = 1.1600 \times 10^{-3} + 4.373 \times 10^{-6} \quad (20)$$

$$a_\mu^{(\text{corr})} = 1.1644 \times 10^{-3} \quad (21)$$

- T0 (corrected): $a_\mu = 1.1644 \times 10^{-3}$
- Experiment: $a_\mu = 1.1659 \times 10^{-3}$
- Deviation: **0.13%**

Tau: More Complex Fractal Structure

Formula:

$$a_\tau = a_e + \frac{4\pi}{f p_\tau} \quad (22)$$

where:

- $p_\tau = 4/3$: Stronger fractal branching

Meaning of $p_\tau = 4/3$:

This is the Hausdorff dimension of the frontier of planar Brownian motion and of the planar self-avoiding walk. (*Corrected on 30 Sept. 2026: previously "box-counting dimension of many fractals (e.g., Koch curve, Mandelbrot set)" – the Koch curve has $\log 4 / \log 3 = 1.262$, the boundary of the Mandelbrot set 2.*)

Numerical Calculation:

$$\Delta a_{\text{fractal}} = \frac{4\pi}{7500^{4/3}} = 8.560 \times 10^{-5} \quad (23)$$

$$a_\tau = 1.184 \times 10^{-3} + 8.560 \times 10^{-5} \quad (24)$$

$$a_\tau = 1.269 \times 10^{-3} \quad (25)$$

Status: This is a **prediction** – tau-g-2 has not been measured yet.

Corrected prediction:

$$a_{\tau}^{(\text{corr})} = a_e^{(\text{corr})} + \Delta a_{\text{fractal}} = 1.1600 \times 10^{-3} + 8.560 \times 10^{-5} \quad (26)$$

$$a_{\tau}^{(\text{corr})} = 1.2456 \times 10^{-3} \quad (27)$$

.4 Two Classes of Predictions: Absolute Values vs. Ratios

Why 2% Deviation for Absolute Values?

Where do the 2% in the ideal formula come from?

The 2% deviation of the ideal formula (with $k_{\text{geom}} = 2.224$) can be traced back to the fractal correction:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{(1 - 100\xi)^{3/2}} = \frac{2.2236}{0.98007} = 2.2688 \quad (28)$$

Physical meaning:

- $K_{\text{frak}} = 1 - 100\xi = 0.98667$ is the universal fractal correction of T0 theory
- The exponent $3/2 = D/2$ (half spatial dimension) describes the effect of fractal structure on the 4D→3D projection
- This correction is **not a fit**: K_{frak} follows directly from ξ and already appears in the mass formulas; the exponent $3/2$ is a geometrically motivated choice
- The experimentally reconstructed value $k_{\text{rec}} = 2.2696$ agrees with the correction to 0.033%

Consistency: The same fractal correction K_{frak} appears in:

1. Particle masses: $m_i = r_i \cdot \xi^{p_i} \cdot v \cdot K_{\text{frak}}$
2. Gravitational constant: $G = f(\xi, K_{\text{frak}})$
3. g-2 anomalies: $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$

This gives a unified picture: **one** correction K_{frak} following from ξ , with **different** exponents depending on the physical context.

Ratios are Mathematically Exact

In contrast to absolute values, **ratios of differences** are structurally exact:

$$\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = \frac{4\pi/f^{4/3} - 4\pi/f^{5/3}}{4\pi/f^{5/3}} = f^{1/3} - 1 \quad (29)$$

Why is this exact?

- The common factor 4π cancels out
- The projection factor k_{geom} cancels out
- Only the fractal exponents ($5/3$ and $4/3$) determine the ratio
- The result depends **only** on f : $f^{1/3} - 1 = 18.57$

Important

Fundamental Distinction **Absolute values:**

- Depend on k_{geom} , f , and SI conversion

- 2% deviation due to higher-order quantum effects
- Consistent with all T0 predictions

Ratios:

- Depend **only** on f
- k_{geom} and SI factors cancel out
- Mathematically exact from fractal exponents
- Difference $< 10^{-13}$ (numerical precision)

⇒ Within the model, the ratio prediction is **not an approximation**, but an **exact geometric relation**.

Analogy to the Koide Formula

This behavior is analogous to the Koide formula for lepton masses:

- **Individual masses:** 0.4–1.2% deviation
- **Koide ratio:** $\pm 0.0004\%$ precision

The ratio is **more robust** than absolute values because systematic factors cancel out.

For g-2 in T0:

- **Absolute values:** 2% deviation
- **Ratio** $\Delta a(\tau - \mu)/\Delta a(\mu - e)$: Exactly $= f^{1/3} - 1$

The difference between the two classes reflects the **geometric structure** of the theory.

.5 Precise Ratio Predictions**Analogy to the Koide Formula**

The Koide formula for lepton masses:

$$\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \pm 0.0004\% \quad (30)$$

shows: **Ratios** are more precise than absolute values.

Question: Does this also hold for g-2?

The Ratio of Differences

Define the differences:

$$\Delta a(\mu - e) = a_\mu - a_e = \frac{4\pi}{f^{5/3}} \quad (31)$$

$$\Delta a(\tau - \mu) = a_\tau - a_\mu = \frac{4\pi}{f^{4/3}} - \frac{4\pi}{f^{5/3}} \quad (32)$$

Ratio:

$$\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = \frac{4\pi/f^{4/3} - 4\pi/f^{5/3}}{4\pi/f^{5/3}} \quad (33)$$

$$= \frac{f^{5/3}}{f^{4/3}} - 1 \quad (34)$$

$$= f^{5/3-4/3} - 1 \quad (35)$$

$$= f^{1/3} - 1 \quad (36)$$

Important

Core Prediction

$$\boxed{\frac{\Delta a(\tau - \mu)}{\Delta a(\mu - e)} = f^{1/3} - 1 = 18.57} \quad (37)$$

This relation is:

- **Depends only on** f (no further parameter)
- **Independent** of k_{geom}
- **Exact** (difference $< 10^{-13}$)
- **Testable** at Belle II

Numerical Verification

With $f = 7500$:

$$f^{1/3} = 7500^{1/3} = 19.57 \quad (38)$$

$$f^{1/3} - 1 = 18.57 \quad (39)$$

From T0 values:

$$\Delta a(\mu - e) = 4.373 \times 10^{-6} \quad (40)$$

$$\Delta a(\tau - \mu) = 8.123 \times 10^{-5} \quad (41)$$

$$\text{Ratio} = \frac{8.123 \times 10^{-5}}{4.373 \times 10^{-6}} = 18.57 \quad (42)$$

Agreement: identical, as expected algebraically.

Testable Prediction for Tau

With experimental values for e and μ :

$$a_e^{\text{exp}} = 1.160 \times 10^{-3} \quad (43)$$

$$a_\mu^{\text{exp}} = 1.166 \times 10^{-3} \quad (44)$$

$$\Delta a(\mu - e)^{\text{exp}} = 6.000 \times 10^{-6} \quad (45)$$

Prediction:

$$\Delta a(\tau - \mu) = \Delta a(\mu - e)^{\text{exp}} \times (f^{1/3} - 1) \quad (46)$$

$$= 6.000 \times 10^{-6} \times 18.57 \quad (47)$$

$$= 1.114 \times 10^{-4} \quad (48)$$

$$a_\tau^{\text{predicted}} = 1.166 \times 10^{-3} + 1.114 \times 10^{-4} \quad (49)$$

$$= 1.277 \times 10^{-3} \quad (50)$$

(Note of 30 Sept. 2026: calculated correctly, but with the measured values rounded to three digits, 1.160 and 1.166; with the exact values of a_e and a_μ , 1.282×10^{-3} follows (Rev. 12, Doc. 158) – a difference in precision, not a calculation error.)

.6 The Fractal Correction $K_{\text{frak}}^{3/2}$: Resolving the 2% Deviation

Central Observation

The 2% deviation of the ideal g-2 formula **corresponds** to the fractal correction $K_{\text{frak}}^{3/2}$, already known from the mass formulas:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{(2/\sqrt{\varphi})\sqrt{2}}{(1 - 100\xi)^{3/2}} = \frac{2.2236}{0.98007} = 2.2688 \quad (51)$$

The experimentally reconstructed value $k_{\text{rec}} = 2.269\%$ agrees with this to **0.033%**.

Why the Exponent $3/2$?

The exponent $3/2 = D/2$ is justified geometrically as follows:

- The fractal correction $K_{\text{frak}} = 1 - 100\xi$ describes the deviation of the real lattice from the ideal crystal structure
- In the 4D→3D projection (k_{geom}), this correction acts across $D = 3$ spatial dimensions
- Per dimension: $K_{\text{frak}}^{1/2}$ (square root of the 1D correction)
- Total: $K_{\text{frak}}^{D/2} = K_{\text{frak}}^{3/2}$

Consistency with Other T0 Corrections

Physical Quantity	K_{frak} Exponent	Interpretation
Particle masses	K_{frak}^1	Scalar correction (1D)
g-2 projection factor	$K_{\text{frak}}^{3/2}$	3D spatial projection ($D/2$)
Gravitational constant	K_{frak}^1	Scalar correction (1D)

Table 2: Universal fractal correction with context-dependent exponent

The different exponents are **not continuously adjusted parameters** but motivated choices guided by the dimensionality of the respective physical process.

.7 Experimental Tests

Belle II (2027–2028)

Belle II expects sensitivity of $\sim 10^{-6}$ for a_τ .

Test 1: Absolute value

- T0 prediction (ideal): $a_\tau = 1.269 \times 10^{-3}$
- T0 prediction (corrected): $a_\tau = 1.246 \times 10^{-3}$

- From ratio: $a_\tau = 1.277 \times 10^{-3}$

- Difference: 1%

Test 2: Ratio

- T0 prediction: $\Delta a(\tau - \mu)/\Delta a(\mu - e) = 18.57$
- This is the **more precise** prediction
- Independent of absolute calibration

Possible outcomes:

1. **Confirmation:** Ratio ≈ 18.6
 $\Rightarrow$ Supports the fractal structure hypothesis
2. **Deviation:** Ratio $\neq 18.6$
 $\Rightarrow$ Different fractal dimensions or additional physics
3. **Null result:** $a_\tau < 10^{-8}$
 $\Rightarrow$ T0 contributions suppressed or theory needs revision

Fermilab/J-PARC

Further precision improvements for a_μ :

- Reduction of experimental uncertainties
- Clearer determination of SM discrepancy
- Refinement of $\Delta a(\mu - e)$ measurement

.8 Comparison with Other Approaches

Approach	Precision	Parameters	Explainable
QED (SM)	Very high	Many	Yes
T0 (ideal, k_{geom})	2%	0 adjusted	Geometric
T0 (with $K_{\text{frak}}^{3/2}$)	0.03–0.13%	0 adjusted	Geometric

Table 3: Comparison of different approaches

Guideline: Here **explainability** takes priority over precision.

.9 Comparison: Experimental Reconstruction and Fractal Correction

The Central Observation

The ratio $\Delta a(\tau - \mu)/\Delta a(\mu - e) = f^{1/3} - 1$ is **mathematically exact** because the correction factor k_{geom} cancels out completely.

In Rev. 10, the experimentally reconstructed value $k_{\text{rec}} = 2.269$ was considered “semi-empirical.” Now it turns out: this value agrees to **0.033 % with the fractal correction**:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{(1 - 100\xi)^{3/2}} = \frac{2.2236}{0.98007} = 2.2688 \quad (52)$$

Comparison: Reconstructed vs. Fractal Corrected

$$k_{\text{geom}}^{(\text{reconstructed})} = \frac{S_3/f}{a_e^{(\text{exp})}} = \frac{2\pi^2/7500}{1.15965 \times 10^{-3}} = 2.2696 \quad (53)$$

(Corrected on 30 Sept. 2026: previously 1.160×10^{-3} in the denominator – that would give 2.2689; the value 2.2696 belongs to the exact measured value $a_e = 1.15965 \times 10^{-3}$.)

Comparison:

- Geometrically ideal: $k_{\text{geom}} = (2/\sqrt{\varphi}) \times \sqrt{2} = 2.224$
- Fractal corrected: $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2688$
- Reconstructed from experiment: $k_{\text{geom}}^{(\text{rec})} = 2.2696$
- **Difference k_{eff} vs. k_{rec} : 0.033%**

This suggests that the “semi-empirical” reconstruction reflects the **geometric** fractal correction $K_{\text{frak}}^{3/2}$.

Results with Fractal Correction

Lepton	Ideal ($k = 2.224$)	Corr. ($k_{\text{eff}} = 2.269$)	Experiment	Dev.
Electron	1.184×10^{-3}	1.1600×10^{-3}	1.1597×10^{-3}	0.03%
Muon	1.188×10^{-3}	1.1644×10^{-3}	1.1659×10^{-3}	0.13%
Tau	1.269×10^{-3}	1.2456×10^{-3}	(not measured)	Prediction

Table 4: Absolute values with geometric vs. fractal corrected k_{eff}

Important

Crucial Point The 2% deviation of the ideal formula is **largely** explained by $K_{\text{frak}}^{3/2}$:

- Electron: 2.07% → 0.033% (factor 62 improvement)
- Muon: 1.89% → 0.13% (factor 15 improvement) (Corrected on 30 Sept. 2026: previously “2.03% → 0.014% (factor 145)” and “0.15% (factor 13)” – consequence of $k_{\text{eff}} = 2.2688$.)
- The fractal correction is **not a new parameter** but follows from $\xi = 4/30000$ (with the exponent 3/2 as a motivated choice)

Thus the T0 g-2 calculation needs **no parameter adjusted to g-2** (only ξ and f plus the stated choices) and reaches a precision of 0.03–0.13%.

Alternative: Directly from Ratio Relation

Independent of k_{geom} is the calculation directly from the exact ratio:

$$\Delta a(\mu - e)^{(\text{exp})} = a_{\mu}^{(\text{exp})} - a_e^{(\text{exp})} = 6.000 \times 10^{-6} \quad (54)$$

$$\Delta a(\tau - \mu) = \Delta a(\mu - e)^{(\text{exp})} \times (f^{1/3} - 1) \quad (55)$$

$$= 6.000 \times 10^{-6} \times 18.57 = 1.114 \times 10^{-4} \quad (56)$$

$$a_{\tau}^{(\text{Ratio})} = a_{\mu}^{(\text{exp})} + \Delta a(\tau - \mu) \quad (57)$$

$$= 1.166 \times 10^{-3} + 1.114 \times 10^{-4} \quad (58)$$

$$= \boxed{1.277 \times 10^{-3}} \quad (59)$$

Note: This prediction is **independent** of k_{geom} and uses only the exact geometric ratio structure.

Two Complementary Tau Predictions

Method	a_{τ} Prediction	Dependent on
Purely geometric	1.269×10^{-3}	$k_{\text{geom}} = 2.224$ (geometric)
With $K_{\text{frak}}^{3/2}$ correction	1.246×10^{-3}	$k_{\text{eff}} = 2.269$ (geometrically derived)
From ratio	1.277×10^{-3}	Only f (exact)
Range	$1.25\text{--}1.28 \times 10^{-3}$	$\pm 1.5\%$

Table 5: Three T0 predictions for a_{τ}

What does this mean for Belle II?

If Belle II measures:

1. $a_{\tau} \approx 1.28 \times 10^{-3}$:
 - Supports the exact ratio relation $f^{1/3} - 1$
 - Indicates that experimental a_{μ} and ratio structure fit together
 - → **Clearest test in favour of T0 geometry**
2. $a_{\tau} \approx 1.25 \times 10^{-3}$:
 - Supports the fractal correction $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$
 - Supports tracing the 2% deviation back to $K_{\text{frak}}^{3/2}$
 - → **Clearest test in favour of fractal T0 geometry**
3. $a_{\tau} \approx 1.27\text{--}1.28 \times 10^{-3}$:
 - Supports the exact ratio relation $f^{1/3} - 1$
 - ? Fractal correction at tau with different exponent?
4. a_{τ} **outside** $1.24\text{--}1.28$:
 - Not satisfied: T0 structure needs revision

Key Statement

The 2% deviation of the purely geometric T0 predictions is **largely** explained by the fractal correction $K_{\text{frak}}^{3/2}$:

- Electron: 2.07% $\rightarrow$ 0.033%
- Muon: 1.89% $\rightarrow$ 0.13%

The effective projection factor $k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2} = 2.2688$ is **not a new free parameter** but follows directly from $\xi = 4/30000$ (exponent 3/2 as a motivated choice).

The k_{geom} -independent T0 prediction for tau uses the exact ratio relation:

$$a_\tau = 1.277 \times 10^{-3} \quad (60)$$

.10 Important Note: No α in the T0 g-2 Formulas

Note: The T0 formulas for g-2 contain **no** α .

In natural units ($\hbar = c = \alpha = 1$):

$$a_\ell = f(\varphi, \xi, f, \text{generation quantum numbers})$$

The anomalous moment is a **purely geometric quantity**, following from the winding structure in the torsion lattice.

Ratios like $\Delta a(\tau - \mu)/\Delta a(\mu - e) = f^{1/3} - 1$ are **independent** of: • α (fine-structure constant)

- SI conversion factors • k_{geom} (projection factor)

They depend only on the fractal structure.

Further Reading and Resources**T0 Theory and Python Scripts:**

- Repository: github.com/jpascher/T0-Time-Mass-Duality
- Python scripts: github.com/jpascher/T0-Time-Mass-Duality/blob/main/2/python/
- Time-Mass Duality documentation
- Fundamental Fractal Geometric Field Theory (FFGFT)

Experimental Results:

- Fermilab Muon g-2 (2025): muon-g-2.fnal.gov
- Theory Initiative White Paper
- Belle II: www.belle2.org

Related T0 Documents:

- Lepton masses: Systematic derivation from quantum numbers
- Koide formula in T0: Geometric interpretation
- Fractal spacetime: $D_f = 3 - \xi$